

MT 507: Modeling and Simulation in Materials Science

Course Code	MT 507
Course Name	Modeling and Simulation in Materials Science
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Motivation - Introduction to Modeling and Simulation**
 - The need for modeling and simulation techniques in material science
 - Experimental measurements and computational calculation of materials properties
 - The role of modeling in predicting material performance and design.
- **Molecular Dynamics (MD) Simulations**
 - Thermodynamics and statistical mechanics principles and Ensembles
 - Basic principles and algorithms (Verlet, leapfrog)
 - Force fields and potentials for material systems
 - Integration schemes and time step considerations
- **Monte Carlo (MC) Simulations**
 - Importance sampling and the Metropolis algorithm
 - Canonical ensemble MC
 - Applications in materials (e.g., phase transitions)
- **Interatomic Potentials**
 - Lennard-Jones potential, Coulombic interactions
 - Embedded Atom Model (EAM) for metals
 - Force field development and selection for different materials
- **Modeling Phase Transitions**
 - Modeling solid-solid and solid-liquid transitions
 - Nucleation and growth processes
- **Simulation of Specific Material Types**
 - Crystalline materials: Simulation of defects, dislocations, and fracture in metals
 - Nanomaterials: Deformation of nanowires and analysis of grain boundaries
- **Other examples for modelling**
 - Grain Boundary Generation of Al -Dislocation Nucleation in Single Crystal Al
 - Grain Boundary Generation of Cu and Mg

Laboratory Modules

1. NA

Text Books

1. Daan Frenkel, **Understanding molecular simulation.**

References

1. Richard LeSar, **Introduction to Computational Materials Science.**
2. Kurt Binder and Dieter W. Heermann, **Monte Carlo Methods in Statistical Physics.**
3. June Gunn Lee, **Computational Materials Science: An Introduction.**